

Statistical Models for Improved Insurance Pricing and Risk Assessment Using Telematics

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Usage-Based Insurance

- Traditionally auto insurance has relied on data from questionnaires and past driver history (no-claims bonus).
- Premiums vary depending on factors such as gender, age, experience, profession, marital status, vehicle type, credit rating and local crime rates.
- Usage-Based Insurance (UBI) uses telematics data to calculate premiums.
- Benefits of UBI over traditional insurance is it;
 - incentivizes safer driving habits,
 - deters unnecessary driving,
 - can reduce premiums for low mileage and low risk drivers,
 - mitigates adverse selection,
 - eliminates moral hazard and prevents fraud.

Problem

- Pricing models used by insurers are proprietary by nature.
- The public is unable to gauge the exact impact that their driving decisions have on their risk of making a claim, and in turn on their premium price.

We aim to increase transparency around pricing models and produce interpretable results that allow drivers to quantify their exposure to risk.

Data

We used a synthetic data set of driver telematics emulated from a real insurance data set collected by a Canadian-owned insurance co-operative under a UBI program (So et al., 2021).

References

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Methodology

- Clustering**
Using Gaussian mixture modelling we cluster policyholders with similar driving profiles, which translates to identifying groupings of policyholders with elevated and reduced probabilities of making a claim.
- Regressions**
We fit cluster-specific logistic regression models to predict the probability of making a claim for each policyholder. Each cluster may exhibit different relationships with the feature variables, and fitting separate logistic regressions within each cluster allows for capturing these variations.

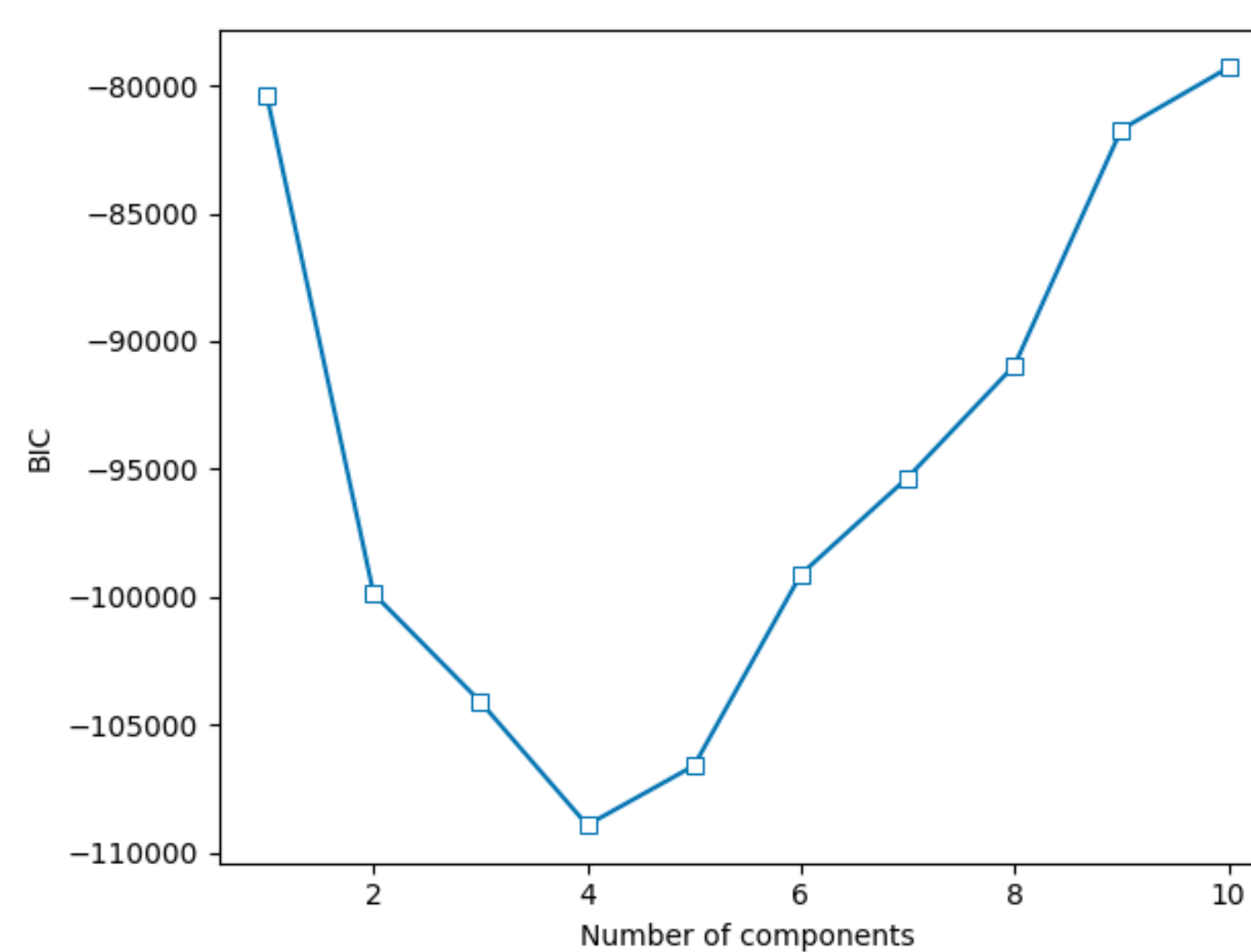
Results

To choose the number of components and type of covariance structure in the Gaussian mixture model the BIC was used. Figure (a) shows the BIC for the number of components ranging from 1 to 10 for a model with a full covariance type, i.e. each component has its own general covariance matrix. Cluster 0 consisted of low-risk policyholders as it has a claim rate of 0.9% and accounted for 18% of the data set. The other clusters had a claim rate between 4.48% and 5.13%. Compared to the average of the other clusters it had the oldest policyholders with the highest credit scores, longest years with a no claim bonus, lowest annual mileage, and lowest sudden acceleration and sudden brakes per 1000 miles.

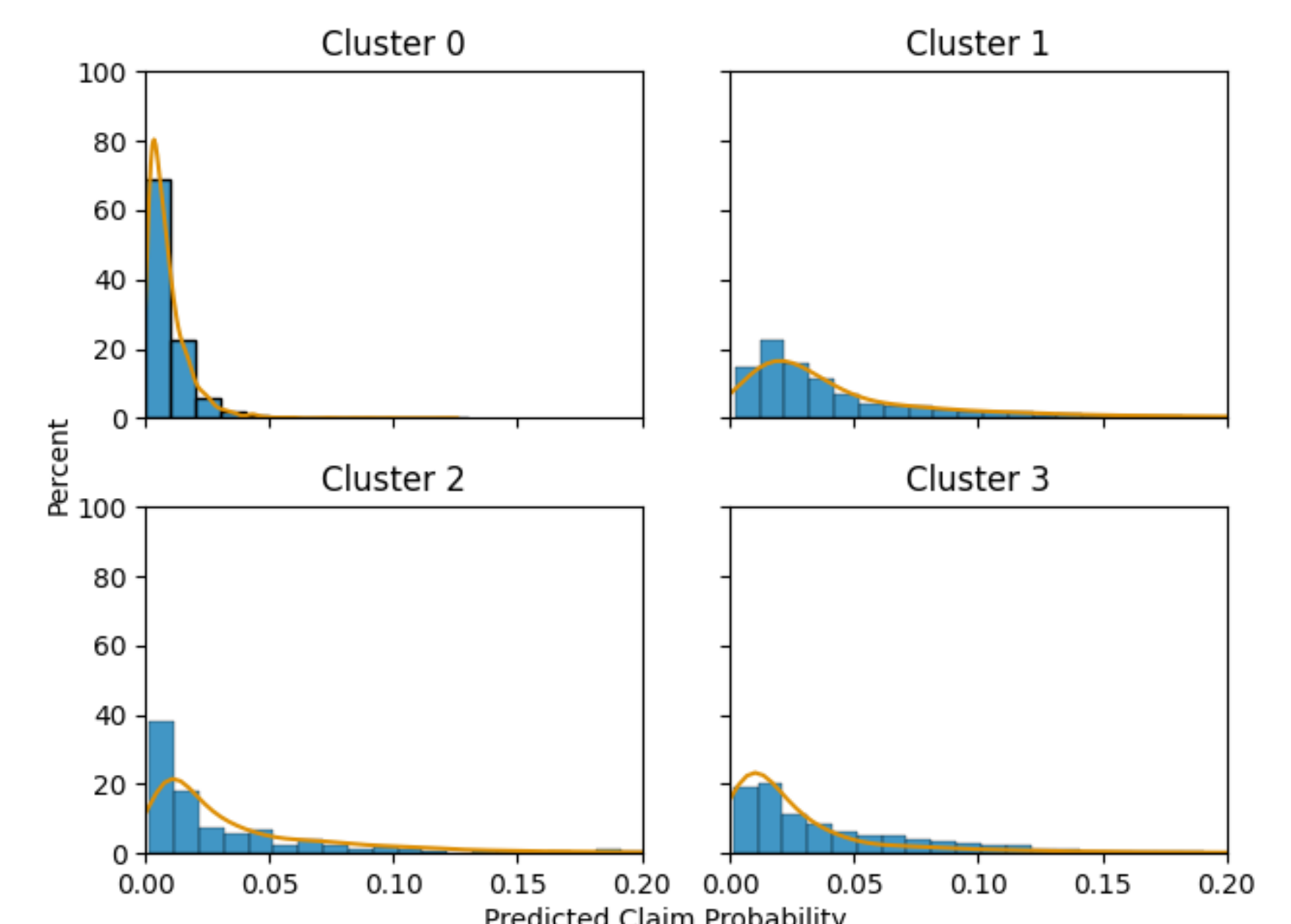
Table 1

	Cluster 0	Cluster 1	Cluster 2	Cluster 3
N	18080	25279	1563	55078
Claims	178	1296	70	2728

In each regression the annual mileage and total number of days driven are significant variables with positive relationships with the probability of a claim. For cluster 0, an increase in sudden accelerations per 1000 miles increased claim likelihood whereas the opposite effect is seen for cluster 1 and 3. Cluster 1 and 3 had the same variables but with different coefficient values.



(a) BIC for Gaussian mixture model where each component has its own general covariance matrix (EVV in **mclust** nomenclature).



(b) Histogram of predicted claim probabilities for observations in each cluster.

Future Research

Future research will incorporate vehicle diagnostics into modelling of claim frequency and claim severity. This extends the current knowledge of driving behaviour and usage patterns to a vehicle's health. The use of vehicle diagnostics in claim prediction models can then be used for proactive maintenance, i.e. early detection and intervention to reduce the number of future claims.

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